Predicting Loan Default

Using Machine Learning

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The Problem

Loan defaults create financial losses for lenders, making **effective credit risk assessment** essential.

Can machine learning identify borrowers at higher risk of default?

What factors are most associated with loan default risk?

Factors That May Help Identify Loan Default

Loan Factors

- Loan Amount
- Installment
- Interest Rate
- Loan Term

Borrower Profile

- Annual Income
- Debt-to-Income
- Employment Length

Credit History

- Credit Grade
- Open Accounts
- Revolving Accounts
- Delinquencies

Account History

- Loan Age
- Inquiries
- Recent Activity

Data Information

Dataset & Source

Source Platform

Kaggle

Dataset Name

Lending Club Loan

Timeline

June 2007 - Dec 2018

Key Metrics

2.26M+

Original Loan Records

145

Original Features

75,000

Modeling Sample

Target Variable

Binary classification target classes representing repayment outcome:

0 Non-Default

Fully Paid / Current

1 Default

Charged Off / Default

Target Variable Definition

Binary Classification Problem

0–Non-Default	1–Default
• Fully Paid • Current • In Grace Period • Late (16-30 days) • Late (31-120 days) • etc	• Charged Off • Default • Does not meet the credit policy. Status: Charged Off

- This project uses supervised machine learning to solve a binary classification problem, in which the original ‘`loan_status`’ variable was grouped into two target classes based on loan repayment outcomes.
- The original ‘`loan_status`’ variable was removed after target creation to prevent data leakage.

Data Preparation & Feature Transformation

01

Data Cleaning & Refinement

Removed irrelevant variables and columns with excessive missing values to clean the baseline dataset.

02

Leakage Prevention

Removed variables containing information unavailable at predi ction time (e.g., `debt_settlement_flag`)

03

Missing Value Treatment

Imputed missing values and created missing-value indicator features where appropriate.

04

Feature Encoding

Converted dates into numerical features (e.g., `loan_age_years`), encoded categorical variables, and created derived features (e.g., `grade_combined`)

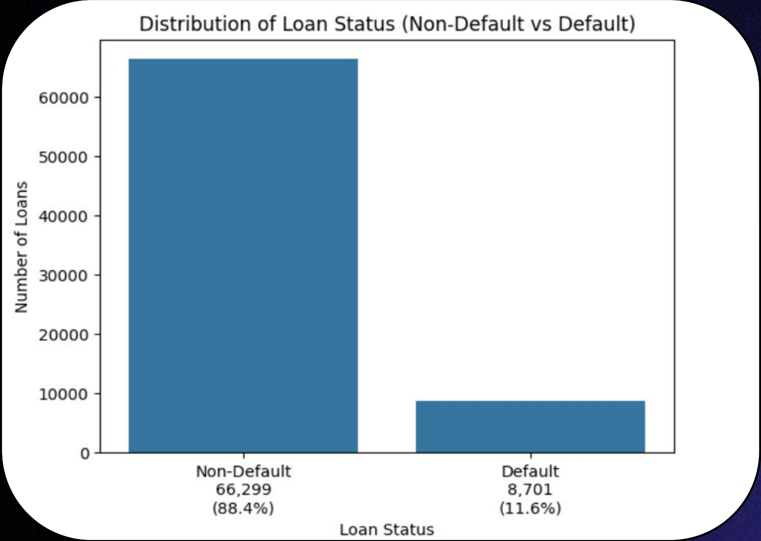
05

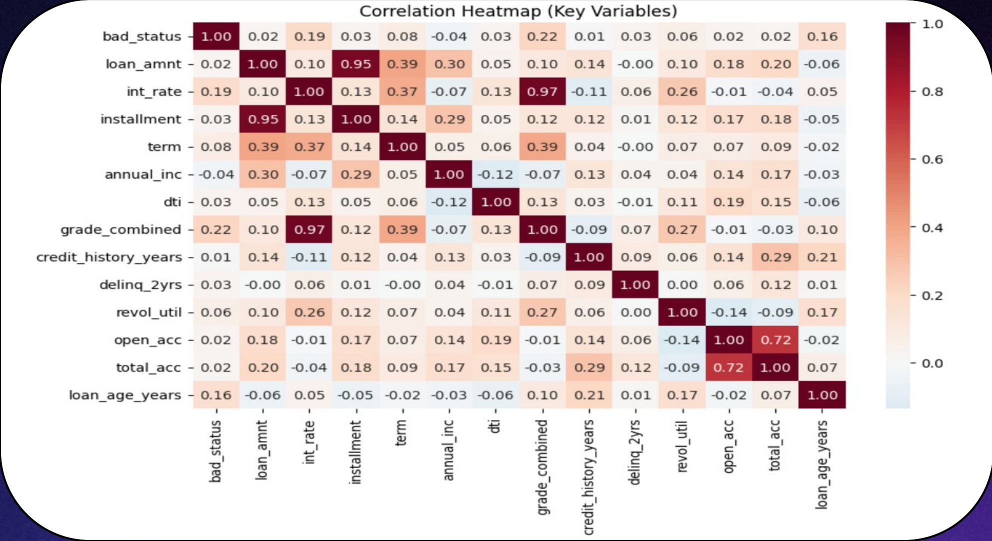
Model Preparation

Performed standard 80/20 train-test split and applied feature scaling for Logistic Regression.

Exploratory Data Analysis

Target & Correlations





Class Imbalance

Significant imbalance: **88.4% non-default** vs. **11.6% default**.

Addressed during modeling using class weighting and SMOTE.

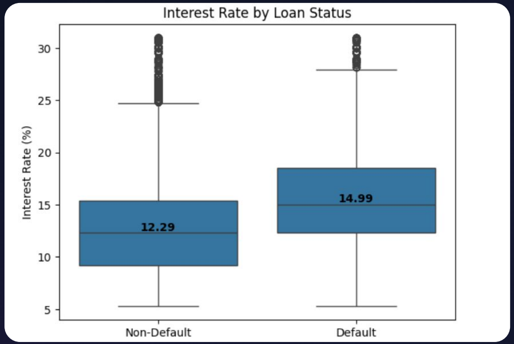
Key Correlation Highlights

grade_combined had the strongest correlation with default status (**0.22**), followed closely by **interest_rate** (**0.19**).

Loan grade & interest rate are highly correlated (**0.97**), indicating strong multicollinearity.

Exploratory Data Analysis:

Borrower Risk Characteristics

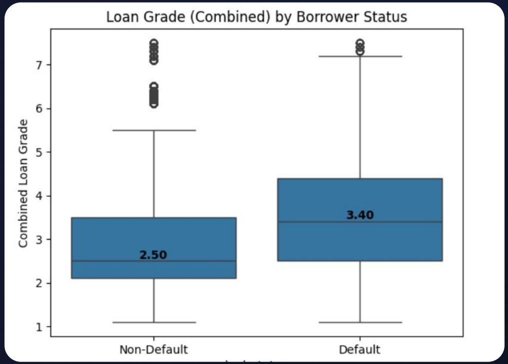


Interest Rate

14.99% (Default)

vs. 12.29% (Non-Default)

Defaulted loans tended to have significantly higher median interest rates.

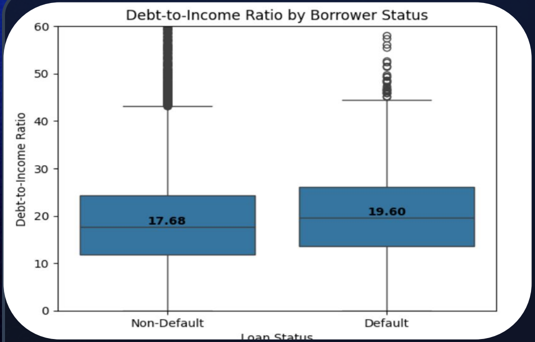


Loan Grade

3.40 (Default)

vs. 2.50 (Non-Default)

Defaulted loans tended to have higher-risk (larger numerical) loan grades.



Debt-to-Income

19.60% (Default)

vs. 17.68% (Non-Default)

Defaulted loans had slightly higher DTI ratios, though the difference was relatively modest.

Model Development & Optimization

Model Families & Candidate Variants

Logistic Regression

Tested 5 variants to establish a robust linear baseline:

- **Baseline LR**
- **Balanced LR**
- **Lasso Regularization**
- **LR with SMOTE**
- **Tuned LR**

Random Forest

Ensemble decision trees to handle nonlinear structures:

- **Baseline RF**
- **RF with SMOTE**
- **Tuned RF**

XGBoost

Advanced gradient boosting for maximum predictive power:

- **Baseline XGBoost**
- **Tuned XGBoost**

Model Optimization Strategies

01 Class Weighting

Assigned greater cost/weight to the minority default class during model training to penalize misclassifications.

02 SMOTE Oversampling

Created synthetic minority-class samples exclusively in the training partition to naturally address severe class imbalance.

03 Hyperparameter Tuning

Tested wide parameter grids (regularization strength, tree depth, and learning rates) to optimize generalization performance.

Model Evaluation & Results

MODEL COMPARISON

Performance Summary of Five Models

★ Higher is better for all metrics

Evaluation Goal: Balance strong predictive performance with the ability to effectively identify defaulted loans.

Key Takeaway: Each model offers a different trade-off between accurately identifying defaults and maintaining high overall accuracy.

METRIC	Balanced Logistic Regression	Tuned Logistic Regression	Tuned Random Forest	Baseline XGBoost	Tuned XGBoost
Accuracy	70.0%	69.0%	66.0%	69.0%	**71.0%**
Recall – Non-Default (Class 0)	70.0%	69.0%	65.0%	69.0%	**71.0%**
Recall – Default (Class 1)	68.0%	69.0%	**73.0%**	71.0%	67.0%
F1-Score: Default (Class 1)	34.0%	34.0%	33.0%	**35.0%**	**35.0%**
ROC-AUC	75.9%	75.8%	75.8%	**77.2%**	77.0%
Gini Coefficient	0.518	0.516	0.516	**0.543**	0.539

Model Insights

Logistic Regression (Balanced/Tuned): Provides consistent, stable performance across both classes.

Tuned Random Forest: Achieves the highest default recall (73%) but lower overall accuracy and the lowest non-default recall.

Baseline XGBoost: Offers the highest Gini (0.543) and ROC-AUC (77.2%), with strong default recall (71%).

Tuned XGBoost: Achieves highest accuracy (71%) and non-default recall (71%).

Accuracy
Overall % of correctly classified loans; can be misleading with imbalanced data.

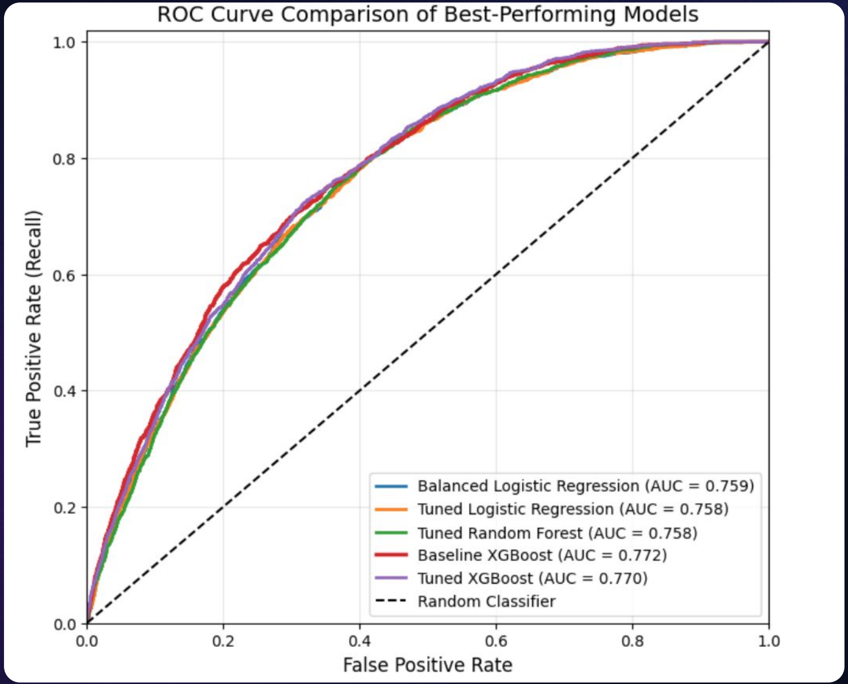
Recall (Default)
Percentage of actual defaults correctly identified — key metric for credit risk.

F1-Score (Default)
Balances precision and recall specifically for the default class.

ROC-AUC
Measures overall model ability to separate default and non-default.

Gini
Measures discriminatory power; mathematically derived from ROC-AUC.

ROC Curve Comparison of Best Performing Models



Performance Overview

All models demonstrate similar and solid discriminatory performance, with ROC-AUC ranging from 0.758 to 0.772.

FINAL CHOICE

Baseline XGBoost

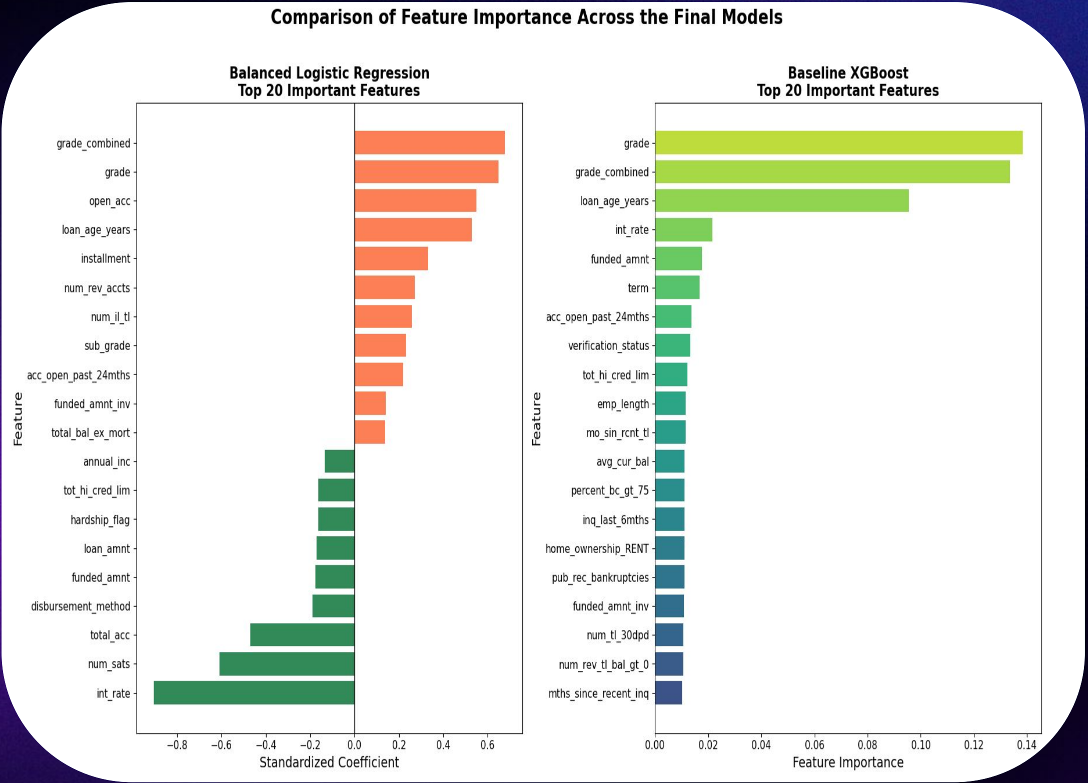
Achieves the highest ROC-AUC (**0.772**) and Gini coefficient (**0.543**), with strong default recall (71%).

Balanced Logistic Regression

Remains a strong, highly interpretable alternative with an ROC-AUC of 0.759.

Final model choice depends on business priorities (recall vs. interpretability).

Feature Importance and Key Predictors



Key Predictors loan grade and loan age emerged as important predictors across both models.

Logistic Regression provides directional relationships, but multicollinearity among correlated features may affect coefficient magnitude and sign (e.g., interest rate).

XGBoost Model captures more complex relationships and interactions, but feature importance does not indicate direction.

Despite different approaches, both models identify similar **core risk factors**

Conclusions & Recommendations

Main Findings

- Baseline **XGBoost** was selected as the final model, achieving the highest ROC-AUC (0.772) and Gini (0.543).
- **Logistic Regression** remains a strong, highly interpretable alternative.
- Results highlight key trade-offs between default detection, predictive power, and model interpretability.

Scope & Application

- Evaluates default risk among **already originated** loans, rather than making front-end decisions.
- Supports ongoing **portfolio risk assessment** by identifying key risk patterns and characteristics.

Future Work

- Evaluate the model on the full LendingClub dataset.
- Validate performance using out-of-time data.
- Explore additional boosting algorithms (LightGBM, CatBoost).
- Optimize classification threshold based on business risk priorities.
- Apply SHAP for model interpretability.